OpenCaseLaw: An Open, Verifiable, Multilingual Legal Knowledge Graph for Switzerland

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Abstract

We present **OpenCaseLaw**, an open, continuously-refreshed corpus of **971,437** Swiss court decisions across **108 courts** and **28 cantons plus federal**, in German, French, and Italian, paired with a citation graph of **8.64M** edges (**6.50M** resolved, 75.2 %) and a statute graph of **11.26M** edges over **283,052** distinct provisions. The corpus is bridged to **5,516** federal and **15,722** cantonal laws (**753,842** articles in total), **362** CC-BY scholarly commentaries, and **5,292** Federal Council Botschaften (Materialien, DE/FR/IT-parallel, with **381,711** full-text paragraphs and **8,124** article-anchored links). The infrastructure is served through a deployed MCP/REST endpoint at `mcp.opencaselaw.ch` (31 tools), a Word-add-in client, and a public dashboard. We use the resource to introduce three contributions: (i) a first cross-lingual leading-case retrieval benchmark for Swiss law, with author-extracted parallel queries in DE/FR/IT against 50 most-cited decisions, where we measure directional asymmetries (overall MRR@10=0.630; IT→DE=0.776; DE→FR=0.333). We disclose a coupling between query construction and target regestes that limits the realism of the benchmark and discuss it in Section 11; (ii) a five-rail closing audit pipeline for legal-RAG output, deployed in production, that decomposes hallucination by named error class (case-citation existence, statute resolution, verbatim-quote, decision-date sanity, and an opt-in proposition-grounding judge); and (iii) a four-layer dataset health framework with 61 codified checks, drift detection, and a publish-gate that blocks production updates on critical regression. The corpus, citation graph, audit pipeline, and benchmark are released under permissive licences (CC-BY-4.0 / MIT / CC0) with a public dashboard.

1 Introduction

Legal artificial intelligence has converged on retrieval-augmented generation (RAG) as the standard architecture for question answering over jurisprudence. Yet recent measurement studies report fabrication rates of 58–82 % on general-purpose LLMs and 17–33 % on commercial legal-RAG products [Dahl et al.(2024)Dahl, Magesh, Suzgun, and Ho, Magesh et al.(2025)Magesh, Surani, Dahl, Su
The mitigation literature is methodologically sophisticated but concentrated on *English-language, single-jurisdiction* settings where retrieval substrates and ground-truth citation graphs are limited.

Switzerland is the most multilingual major civil-law jurisdiction. The authoritative version of a leading Swiss federal decision may be in German, French, or Italian, and citations cross language boundaries: in our resolved citation graph, **35.6 % of resolved citations are cross-lingual** (Table 3). Building a credible RAG system for Swiss law therefore requires (a) a corpus that spans the federal–cantonal landscape across all three languages, (b) a citation and statute graph that can verify generated claims, and (c) evaluation that measures cross-lingual retrieval as a first-class concern.

We present **OpenCaseLaw**, an open infrastructure that fills these gaps. We make three contributions:

C1 — Resource. We release the largest open, multilingual, federation-complete corpus of Swiss court decisions that we are aware of, with same-day refresh of federal Supreme Court decisions (poller-driven, ∼15 min latency) and nightly refresh of all other federations, a knowledge graph spanning citations, statute references, and Federal Council Botschaften (*Materialien*, the primary source for teleological interpretation in Swiss civil law), and a deployed MCP/REST infrastructure with 31 retrieval and analysis tools.

C2 — Benchmark. We introduce a *cross-lingual leading-case retrieval* task. Our headline result (Section 7) shows that the BM25+RRF baseline scores MRR@10=0.630 averaged across 150 queries (50 leading cases×3 languages), with directional asymmetry: IT→DE retrieves at MRR=0.776, exceeding DE→DE on the same targets (0.617), while DE→FR is the hardest cell at 0.333. We disclose that queries are extracted from each language's official regeste, which makes the absolute numbers an upper bound on realistic legal-research queries (Section 11).

C3 — Verification. We deploy and evaluate a five-rail closing audit pipeline (Section 6) that decomposes legal-RAG hallucination into named error classes, plus a four-layer dataset-health framework (Section 9) that codifies 61 dataset invariants with drift detection. The framework blocks the production publish on a critical regression.

2 Related work

Open legal corpora. The Caselaw Access Project [Harvard Law School Library(2018)] releases 6.7M US decisions but does not publish a citation graph or maintain a retrieval system. The *Open Legal Data* platform [Ostendorff et al.(2020)] (laws and decisions, OLD36k initial release; OLD201k extension by Milz et al. 2025 grows to 201 k decisions) and *LeC-Net* [Anonymous(2025a)] provide German legal citation networks but no statute graph and no multilingual coverage. *Pile of Law* [Henderson et al.(2022)Henderson, Krass, Zheng, Guha, Manning, Jurafsky, and *MultiLegalPile* [Niklaus et al.(2023)] are pretraining corpora rather than retrieval substrates. Within Swiss legal NLP, *Swiss-Judgment-Prediction* [Niklaus et al.(2021)Niklaus, Chalkidis, and Stürmer] and *SCALE* [Rasiah et al.(2023)] address narrow tasks (one court, multi-task NLP); we instead release the underlying resource and target retrieval and verification.

Legal-RAG benchmarks. *LegalBench-RAG* [Pipitone and Houir Alami(2024)] introduces retrieval evaluation for US legal RAG. *Butler & Butler 2026* [Butler and Butler(2026)] adds end-to-end correctness×groundedness×retrieval-accuracy decomposition on Victorian criminal law. *LegalRAG* [Anonymous(2025b)] proposes a multilingual hybrid pipeline. *HousingQA / BarExamQA* [Zheng et al.(2025)Zheng, Guha, Arifov, Zhang, Skreta, Manning, Henderson, and Ho] are US-focused multiple-choice. None of these benchmarks measure cross-lingual retrieval where the target authoritative version may be in a different language than the query.

Hallucination measurement and mitigation. *Dahl et al. 2024* [Dahl et al.(2024)Dahl, Magesh, Suzgun, a and *Magesh et al. 2024* [Magesh et al.(2025)Magesh, Surani, Dahl, Suzgun, Manning, and Ho] establish hallucination rates on general LLMs and commercial legal-RAG. Self-RAG [Asai et al.(2023)Asai, Wu, and CRAG [Yan et al.(2024)Yan, Gu, Zhu, and Ling] propose verification-by-judge and corrective retrieval, both as black-box LLM passes. We instead decompose verification into per-error-class deterministic rails plus an opt-in judge.

Table 1: OpenCaseLaw corpus overview. Snapshot 2026-05-08.

Quantity	Value
Total decisions	971,437
Distinct courts	108
Cantons + federal	28
Date range	1875–2026
German (de)	449,575 (46.3%)
French (fr)	441,158 (45.4%)
Italian (it)	80,704 (8.3%)

Positioning. OpenCaseLaw is, to our knowledge, the first project that combines (a) a multilingual federation-complete legal corpus, (b) a citation and statute graph at this scale, (c) a cross-lingual leading-case benchmark, and (d) a deployed verification pipeline, in a single open release.

3 Resource: corpus

The OpenCaseLaw corpus comprises **971,437 decisions** from **108 distinct courts** across **28 cantons plus federal**. The federal Supreme Court (BGer) is refreshed within approximately 15 minutes of its document-service publication via a dedicated poller; all other federations (BVGer, BStGer, BPatGer, regulatory authorities, 53 cantonal sources, ECHR-Swiss-subset, historical archives) are refreshed by a nightly publish at 03:00 UTC. Coverage spans **1875–2026** and three official languages: German (449,575 / 46.3 %), French (441,158 / 45.4 %), Italian (80,704 / 8.3 %). The federation aggregates **54 individual scrapers** ranging from federal courts (BGer, BVGer, BStGer, BPatGer) to per-canton portals to the European Court of Human Rights (Swiss subset; see below), historical archives (BGE 1875–1953, MKG 1915–2025), and regulatory authorities (FINMA, WEKO, EDÖB, UBI, ElCom, PostCom, ComCom).

ECHR Swiss subset. The Swiss-relevant subset of European Court of Human Rights jurisprudence (**2,788 decisions**: 1,153 chamber + 93 grand-chamber + 230 committee + 835 HUDOC entries + 477 BGE-EGMR mirror) is integrated as a first-class federation. The set is refreshed daily; on the snapshot date the most recent ingest carried a same-day Article 11 violation against Switzerland (application 30781/22, decided 2026-05-07). The subset matters because **Article 6 ECHR** is the second-most-cited statutory provision in our graph (after federal-court procedure), and the official versions of ECHR-Switzerland judgments are typically French and English, requiring a multilingual retrieval substrate.

Completeness. A persistent challenge for Swiss legal corpora is that several cantonal portals geo-block or rate-limit research scrapers. We document and release the recovery techniques in our reproducibility manifest, including a reverse-SOCKS tunnel via residential IP for Hetzner-blocked portals (JU, NE) and Wayback Machine recovery for closed historical archives (BL, ∼903 ALS judgments).

Licensing. The base corpus is released under CC-BY-4.0; commentaries from OnlineKommentar.ch are CC-BY; from OpenLegalCommentary.ch CC-BY-SA. Decision IDs are stable across daily refreshes.

Table 2: Citation graph. Edges extracted from federal-court decisions; resolved means the cited target is identified in the corpus.

Quantity	**Value**
Decisions analysed for outgoing citations	971,377
Citation edges (source $\rightarrow$ target reference)	8,643,604
Edges with target resolved to a corpus decision	6,496,170 (75.2%)
Distinct decisions cited at least once	210,895
Decisions cited $\geq$ 100 times	8,601
Decisions cited $\geq$ 1,000 times	888

Table 3: Cross-lingual citation matrix. Rows are source decision languages; columns are cited (target) decision languages. Cells give resolved citation edges. Off-diagonal = 2,309,722 / 6,496,170 = 35.6% of all resolved citations.

	$\rightarrow$ DE	**$\rightarrow$ FR**	**$\rightarrow$ IT**
DE $\rightarrow$	2,161,704	374,869	25,214
FR $\rightarrow$	1,656,008	1,989,062	52,123
IT $\rightarrow$	128,707	72,801	35,682

4 Resource: knowledge graph

Citation graph. We extract citation edges by full-text scanning every decision in the corpus (**971,377** decisions in the reference-graph snapshot; the small lag versus the corpus size in Table 1 reflects that the citation-graph rebuild runs after the FTS5 build, so the most recent few hundred decisions enter the citation graph the following nightly), recovering **8,643,604 outgoing references** of which **6,496,170 (75.2 %)** resolve to a target decision in the corpus by exact-match on docket normalization. We empirically decompose the remaining 2.19 M unresolved mentions (24.8 %) by mechanism (`benchmarks/citation_resolution_analysis.py`): **44.6 %** (978,413 mentions) are docket-normalization drift in 2007–2009 *Bundesgericht* references where the corpus stores the docket as "`8C 862_2008`" (with a space) while the citation extractor produces "`8C_862_2008`" (with an underscore); **27.5 %** (602,803 mentions) are pin-cite failures where Swiss legal citations such as *BGE 125 V 352* pinpoint into the body of a case (page 352 of the case starting at *125 V 351*) rather than naming the case's first page; **25.2 %** (553,064 mentions) are cantonal or lower-court references genuinely outside our current coverage; **2.7 %** (59,163 mentions) are BGE references whose volume/division/page combination we do not hold; **0.05 %** are malformed extraction artefacts. Two tractable fixes — pin-cite-aware lookup (nearest preceding first-page in the same volume + division) and docket-norm space/underscore unification — would lift exact-match resolution to a projected **93.5 %**; we report the deployed 75.2 % as the headline number and the empirical breakdown as a roadmap for v1.1.

The citation graph is heavy-tailed (Table 2): 32 decisions are cited 10,000+ times, 856 are cited 1,000–9,999 times, and the rest follow a power law. The single most-cited decision is BGE 125 V 351 (60,650 incoming citations), the foundational case on the evidentiary weight of party-commissioned medical reports in social-insurance litigation.

Cross-lingual citation. Table 3 reports the language-by-language citation matrix. **35.6 % of all resolved citations are cross-lingual**: French decisions cite German decisions **1.66M** times (FR→DE dominates the off-diagonal), German decisions cite French decisions **0.37M** times, and Italian decisions cite German decisions **129K** times. The asymmetry — DE↔FR is bidirectional but FR→DE outweighs DE→FR by 4.4×, while IT→DE/FR dominates IT→IT — reflects practitioners' practical reliance on German jurisprudence as binding precedent in mixed-language litigation.

Table 4: Statute and doctrinal graph layers. Decision-to-statute edges plus federal/cantonal mirrors plus Materialien (Federal Council Botschaften) and CC-BY commentaries.

Quantity	**Value**
Decision → statute edges	11,255,077
Distinct provisions referenced	283,052
Federal laws (Fedlex SR series)	5,516
Federal articles (DE/FR/IT parallel)	400,405
Cantonal laws (all 26 cantons; 22 direct, 4 LexFind fallback)	15,722
Cantonal articles	353,437
Materialien (Botschaft documents, DE/FR/IT)	5,292
Materialien paragraphs (full-text)	381,711
Article-anchored Materialien links	8,124
Open commentaries (CC-BY)	362

Statute graph. We resolve **11,255,077 decision-to-statute edges** touching **283,052 distinct provisions**. The statute layer is grounded in two parallel mirrors: a federal layer of **5,516 SR-numbered Fedlex laws** (400,405 article rows summed across the parallel DE/FR/IT versions; ∼133 k articles per language) and a cantonal layer of **15,722 laws** (353,437 articles) covering all 26 cantons — 22 via direct portal scraping (LexWork, SIL, ZH OpenData, TI RL; 12,177 laws) plus 4 via LexFind PDF fallback (JU, SZ, UR, VD; 3,545 laws). Top-referenced provisions are dominated by federal-court procedural code: *Bundesgerichtsgesetz* Art. 42 (motivation requirement, ≥200 k references when LTF and BGG abbreviation forms are pooled), Art. 100 (deadline), and *ATSG/LPGA* Art. 61 (social-insurance procedure).

Materialien graph (Federal Council Botschaften). A distinguishing feature of Swiss civil-law interpretation is the *teleological* reading of statutes via *Materialien* — legislative-history documents (Federal Council messages, parliamentary debates) that record legislative intent. We discover Materialien via the Fedlex JOLUX SPARQL endpoint (`typeDocument` = 23) and ingest them as **5,292 Botschaft documents** (DE 1,763, FR 1,765, IT 1,764, ranging 2003–2026), parsed at paragraph granularity (**381,711 paragraphs**) from Akoma Ntoso XML or PDF fallback. We anchor paragraphs to specific `(SR, article)` pairs by traversing the parliament-chain (`parliamentDraftId` → enacted `oc` → codified `cc` → SR), yielding **8,124 article-to-Botschaft links** (DE 2,798, FR 2,742, IT 2,584). The Materialien layer is queryable via `search_materialien` / `get_materialien` MCP tools and surfaces verbatim legislative-intent text under any cited statute article. To our knowledge this is the first open, machine-queryable bridge between the Swiss case-law corpus and the Botschaften that a court would consult in interpretation.

Commentaries. We bridge **362 scholarly commentaries** (CC-BY) from OnlineKommentar.ch and OpenLegalCommentary.ch into the graph, anchored by SR number plus article. Doctrinal layer is currently shallow but growing.

5 Infrastructure

The corpus, graph, statutes, and Materialien are served through four production surfaces:

- **MCP server** (`mcp.opencaselaw.ch/sse`, public SSE). 31 tools partitioned into four families: *retrieval* (`search_decisions`, `get_decision`, `find_leading_cases`, `find_citations`, `find_appeal_chain`, `search_laws`, `get_law`, `search_legislation`, `get_legislation`,

Table 5: Top 15 most-cited decisions in the corpus. Subject area inferred from the decision regeste; in-degree counts citing decisions (with multi-citing pairs counted once per source).

Decision ID	Reporter form	In-degree
`bge_BGE_125_V_351`	BGE 125 V 351	60,650
`bge_BGE_134_V_231`	BGE 134 V 231	35,534
`bge_BGE_122_V_157`	BGE 122 V 157	27,748
`bge_BGE_130_V_343`	BGE 130 V 343	21,778
`bge_BGE_137_V_210`	BGE 137 V 210	20,073
`bge_BGE_135_V_465`	BGE 135 V 465	20,070
`bge_BGE_141_V_281`	BGE 141 V 281	18,760
`bge_BGE_138_III_374`	BGE 138 III 374	18,214
`bge_BGE_126_V_75`	BGE 126 V 75	17,421
`bge_BGE_133_II_249`	BGE 133 II 249	17,053
`bge_BGE_115_V_133`	BGE 115 V 133	16,028
`bge_BGE_125_V_256`	BGE 125 V 256	15,997
`bge_BGE_133_V_108`	BGE 133 V 108	15,962
`bge_BGE_126_V_353`	BGE 126 V 353	15,377
`bge_BGE_125_V_413`	BGE 125 V 413	15,247

`browse_legislation_changes`, `search_commentaries`, `get_commentary`, `search_materialien`, `get_materialien`, `search_practice`, `get_practice`); *analysis* (`analyze_legal_trend`, `get_doctrine`, `get_decision_structure`, `get_erwaegung`, `find_relevant_erwaegung`, `get_regeste`, `get_article_purpose`, `get_case_brief`, `get_statistics`, `list_courts`); *generation* (`generate_exam_question`, `draft_mock_decision`); and *verification* (`check_claim_support` `attest_response`, `cite`). Two additional tools (`update_database`, `check_update_status`) are local-only and hidden by the `REMOTE_MODE` filter on the public deployment.

- **REST API** (FastAPI; export endpoints `.docx / .pdf / .bib / .ris` per decision; OpenAPI spec at `/api/openapi.copilot.json` curated for Microsoft Copilot Studio adoption).
- **Word add-in** (`opencaselaw.ch/word`, public). Lawyer-facing client that surfaces the verification rails as inline citations, with structural client+server PII redaction at boundary.
- **Public dashboards** at https://opencaselaw.ch (corpus stats, multilingual landing) and https://opencaselaw.ch/quality.html (live dataset health).

The retrieval substrate is a SQLite FTS5 index (∼63 GB on disk). The nightly refresh uses an *atomic-rebuild* pattern: the new index is constructed in a sibling `.db.tmp` file, then promoted via `os.replace`; live readers (4 uvicorn workers behind nginx) are never interrupted.

Adoption signal. The MCP endpoint has registered early integration from a Swiss-French law firm (Lalive, via Microsoft Copilot Studio) and from a cantonal-government legal department (État du Valais, custom MCP client), with sustained query volume in legal-research patterns rather than synthetic load. Adoption is descriptive, not an evaluation claim, but it confirms that the deployed stack is exercised by professionals on real research questions.

6 Five-rail closing audit

We deploy a five-rail closing audit on every LLM response generated by the public MCP. Each rail addresses a single named error class: (R1) case-citation existence, (R2) statute reference

Table 6: Top 15 most-referenced statutory provisions. *LTF* (French) and *BGG* (German) refer to the same Bundesgerichtsgesetz / Loi sur le Tribunal fédéral.

Law	**Article**	**Decisions citing**
LTF	Art. 100	192,524
LTF	Art. 42	112,815
LTF	Art. 113	110,490
BGG	Art. 42	100,182
BGG	Art. 66	66,188
LTF	Art. 72	56,916
LPGA	Art. 61	55,429
VWVG	Art. 63	55,327
LTF	Art. 82	52,762
BGG	Art. 106	51,365
ASYLG	Art. 3	51,282
LTF	Art. 74	51,056
CST	Art. 29	48,298
IVG	Art. 28	47,092
BGG	Art. 105	46,018

resolution, (R3) verbatim-quote source matching, (R4) decision-date sanity, and (R5) LLM-judge proposition grounding. R1–R4 are deterministic regex+lookup (∼3 ms total), R5 is judge-mediated (∼2 s, ∼$0.005). The decomposition by error class lets each rail be tested, ablated, and improved in isolation.

The closing audit operates on a draft answer D and a retrieval-state pool S (decisions and statutes the generator had access to). It returns a list of issues $\mathcal{I}(D, S)$, each labelled with one of five named error classes. The five rails operate *in parallel* on the same draft.

Rail 1 — case-citation existence. A regex parses the draft for Swiss case-citation forms (BGE / ATF / DTF, BGer / TF dockets, BVGer / BStGer / BPatGer dockets, MKGE numerical references). Each parsed citation is resolved against the corpus by exact-match lookup with a strict canonical-prefix guard. Latency 2–5 ms regardless of citation count. Catches: invented BGE numbers, mis-typed docket numbers, citations to retracted decisions. An optional pinpoint check requires `E. n.m` or `consid. n.m` markers to match the structured-extraction sidecar (`decision_structure.db`) or the body `regeste` of the cited decision.

Rail 2 — statute-reference resolution. A second regex parses for `Art. N LAW` forms (with German / French / Italian abbreviation synonyms; LTF=BGG, CO=OR, CC=ZGB, etc.) and resolves each against the Fedlex SPARQL mirror cached locally as `statutes.db`. Two-step resolution: abbreviation → SR number → article text, cached at process scope. Latency 1–3 ms. Catches: fabricated law abbreviations, wrong article numbers, references to repealed articles. Connector words ("und", "oder", "et", "ou") are case-sensitively filtered to prevent the regex eating them as fake law codes.

Rail 3 — verbatim-quote source-matching. Quoted substrings of ≥ 30 characters in the draft are extracted (German „...", French « ... », Italian « ... », English curly and ASCII variants) and required to appear verbatim — after whitespace + quote-mark normalisation — in the source pool S. The pool comprises the `regeste` text plus the first 8 k characters of each cited decision, plus the article text of each cited statute. Latency ∼2 ms. Catches: hallucinated quotations, paraphrases inside quotation marks.

Table 7: Search-quality baseline by query language. 100-question golden set, offline (BM25+RRF, no LLM rerank), 2026-03-19. Italian queries score highest because the Italian corpus is the smallest (lowest target ambiguity).

Language	n	MRR@10	Recall@10	Hit@1	nDCG@10
DE	74	0.431	0.422	0.284	0.463
FR	16	0.510	0.630	0.375	0.611
IT	7	0.770	0.893	0.714	0.803

Rail 4 — decision-date sanity. For each verified case citation, parse adjacent `vom` / `du` / `del` DD.MM.YYYY constructs within 60 characters of the citation span and compare to the stored `decision_date`. Latency <1 ms. Catches: model emits a real BGE but pairs it with a wrong date, suggesting confusion between two cases with similar dockets.

Rail 5 — proposition grounding (opt-in, judge-mediated). For each verified citation, a claim sentence is extracted (preceding sentence, with forward-fallback for "In BGE X, ..." patterns). "See"-cite markers (`vgl.`, `cf.`, `siehe`, `cfr.`) suppress the rail. All (claim, source) pairs go to a separate LLM judge call (same model family as the generator) that returns one of {`yes`, `partial`, `no`, `contradicts`, `unrelated`}. Verdicts in {`no`, `contradicts`, `unrelated`} become issues. One judge call per draft regardless of citation count. Latency 1.5–2.5 s. Cost ≈ $0.005 per call.

Output canonicalisation. When all rails pass, the audit emits a `linked_text` field with every parsed citation replaced by its corpus-canonical form wrapped in a Markdown link to `mcp.opencaselaw.ch/entscheid/<id>`. Drafts arrive with messy citation spacing or case and leave publication-ready.

Empirical baseline (from prior-only condition). On the v2 30-question Swiss Legal RAG bench, ∼33 distinct authority citations were emitted across 30 drafts; the deterministic rails (R1–R4) flagged 1 (3 % observed fabrication rate). For context, Magesh et al. [Magesh et al.(2025)Magesh, Surani, D report 17–33 % on commercial legal-RAG and Dahl et al. [Dahl et al.(2024)Dahl, Magesh, Suzgun, and Ho] report 58–82 % on general-purpose LLMs in different settings; we cite these to bracket the literature, not as a like-for-like comparison. The full five-rail audit including the LLM-judge rail R5 reaches a 66.7 % wrong-draft flag rate at a 50 % false-positive rate; calibration of the judge's threshold remains an open problem and the rail should be deployed in advisory mode pending future calibration work.

Retrieval-augmented condition. A retrieval-augmented re-run on the same 30-question bench is the highest-priority follow-on for v1.1 of the benchmark; we publish it as a versioned update on the bench's Hugging Face page when ready.

7 Cross-lingual leading-case retrieval

Setup. We select 50 leading cases from the most-cited subset of the corpus (in-degree range 4,412–53,251, median 8,180), stratified to at most four cases per legal area to ensure diversity across the 22 covered areas (constitutional, social-insurance, federal-court procedure, criminal, civil-procedure, contract, tax, asylum, and more). All 50 cases lie in the top 0.025 % of distinct cited decisions in the corpus by in-degree. For each case we extract one query per language (DE, FR, IT) from terms in that language's official regeste, after dropping article-reference noise (e.g. `Art. 6 Abs. 1 UVG`) and language-specific stop words. The query construction is keyword-list

Table 8: Cross-lingual leading-case retrieval. Cells give MRR@10 (n queries) for retrieval of a leading-case decision when the query is in language Q and the decision's authoritative language is T. 50 leading cases (in-degree range 4,412–53,251, median 8,180; top 0.025 % of cited decisions), with regeste-derived parallel queries in DE/FR/IT. BM25+RRF (no LLM rerank). The 50 cases include 38 originally DE and 12 originally FR; none is originally IT (consistent with the corpus distribution among most-cited decisions). Overall MRR@10=0.630, Hit@10=0.833 across n=150 queries. Numbers are upper-bounded by query–regeste vocabulary coupling; see §11.

Q lang ↓	**T lang →**	**DE**	**FR**	**IT**
DE		0.617 (n=38)	0.333 (n=12)	—
FR		0.566 (n=38)	0.785 (n=12)	—
IT		0.776 (n=38)	0.549 (n=12)	—

Table 9: Cross-lingual leading-case retrieval, Hit@10. Same setup as Table 8. Hit@10 = fraction of queries where the target leading case appears in the top 10 results.

Q lang ↓	**T lang →**	**DE**	**FR**	**IT**
DE		0.789	0.333	—
FR		0.763	1.000	—
IT		1.000	1.000	—

style, not a free natural-language paraphrase; we discuss the realism implications in Section 11. The dataset is released as `benchmarks/swiss_legal_rag_bench/cross_lingual_v1.jsonl` (CC0). Of the 50 cases, 38 are originally German and 12 originally French — consistent with the corpus distribution among most-cited decisions. No selected case is originally Italian; we treat this as a coverage gap (Section 11).

Results. Table 8 reports MRR@10 and Table 9 reports Hit@10 per (Q lang, T lang) cell. Overall MRR@10=0.630, Hit@10=0.833 (125 of 150 queries place the target in the top 10).

Findings.

- **Italian queries retrieve German targets best (MRR=0.776).** Italian queries outperform German queries on the *same* 38 German targets (DE→DE MRR=0.617). One mechanism: Italian regestes carry distinctive shared markers (*DTF*, *cpv.*, *LAINF*) that anchor retrieval, while German queries compete with the broader German subset (449 k decisions, the largest stratum) and so retrieve more in-language distractors.

- **Same-language French is the strongest cell (MRR=0.785, Hit@10=1.0).** For the 12 French-original leading cases, French queries place the target in the top 10 every time and at rank 1 in 67 % of cases.

- **German-to-French is the hardest cell (MRR=0.333, Hit@10=0.333).** German queries struggle to retrieve French-original BGEs because the discriminative German terms in the regeste-derived queries do not appear in the French-language full text and the BM25+RRF baseline does no cross-lingual translation.

- **Diagonal vs. off-diagonal gap is small.** Weighted across n, same-language diagonal cells (DE→DE n=38, FR→FR n=12) score MRR=0.657; the four off-diagonal cells (n=100 total) score MRR=0.616. The 0.04 gap shows that cross-lingual retrieval against a multilingual leading-case index is feasible without translation, although the cell-by-cell variance (0.333 to 0.776) is large.

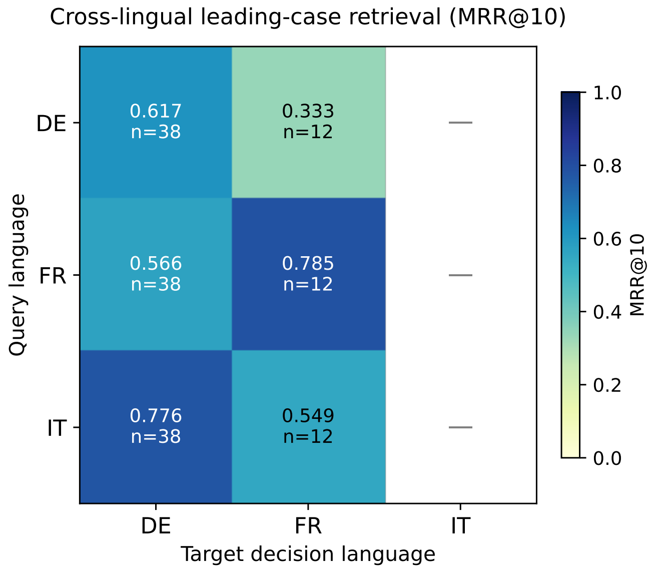

Figure 1: Cross-lingual leading-case retrieval (MRR@10) heatmap. Rows are query languages, columns are target authoritative-version languages.

8 Hallucination stress test

We evaluate the five-rail audit on the v2 30-question Swiss Legal RAG bench [Butler and Butler(2026)], which spans 17 substantive areas of Swiss federal law (Schuldrecht, Strafrecht, Verfassungsrecht, Verfahrensrecht, Mietrecht, Verjährung, ...). Language distribution: DE=18, FR=8, IT=4 (60 / 27 / 13 %). Each question is paired with a corpus-anchored reference answer (statute SR-number plus article number plus language; leading-case `decision_id`(s) where applicable). 16 of 30 questions are anchored on a verified leading-case BGE; 14 carry the `doctrine_from_bge` claim type.

Conditions evaluated. *Prior-only*: the generator answers from training prior alone, no retrieval. We deliberately use the prior-only condition as a worst-case stress test that maximises errors; a retrieval-augmented condition is deferred to v1.1 of the benchmark (Section 11). The audit pipeline is condition-agnostic: the rails make no assumption about whether a draft was produced with or without retrieval, so the same deployment audits both.

Findings (prior-only). Across 30 prior-only drafts containing ∼33 distinct authority citations, the deterministic rails (R1–R4) flagged exactly 1 unresolvable citation: a 3 % *observed* fabrication rate. For comparison, Magesh et al. [Magesh et al.(2025)Magesh, Surani, Dahl, Suzgun, Manning, and report 17–33 % on commercial legal-RAG products and Dahl et al. [Dahl et al.(2024)Dahl, Magesh, Suzgun, and report 58–82 % on raw general-purpose LLMs (different conditions, different jurisdictions, not directly comparable; we cite to bracket the literature). The single failing docket (BGE/DTF 117 II 198) fails consistently across re-runs: with our deterministic rails alone we cannot distinguish a true fabrication from a corpus gap without manual adjudication. The full five-rail audit including the LLM-judge rail reaches a 66.7 % wrong-draft flag rate at a 50 % false-positive rate; calibration of the judge's threshold remains an open problem and the rail should currently be deployed in advisory mode rather than as a hard binary gate.

Retrieval-augmented condition. A side-by-side retrieval-augmented re-run is the highest-priority follow-on for v1.1. We expect retrieval to substantially reduce R1 fabrication-existence flags (the system retrieves real authorities rather than relying on prior); the open question is whether judge-rail false-positive rate increases (more authorities to ground against) or decreases (clearer source pool). We commit to publish the retrieval-augmented numbers as a versioned update on the benchmark's Hugging Face page rather than holding the resource paper's release on it.

9 Dataset health and drift

OpenCaseLaw operates a four-layer quality framework:

L1 Function tests (46 pytest cases, every CI push, <90 s).

L2 Pipeline guards (5 codified `_normalize_*` passes in `build_fts5.py`, every nightly rebuild).

L3 Dataset audit (61 `check_*` functions across 17 axes: schema, dates, dockets, courts, languages, regeste coverage, URL integrity, duplicates, text quality, structure, cross-DB consistency, citation graph, statute graph, floors, per-court drift, exports round-trip, MCP-tool round-trip).

L4 Production smoke (every 5 min via systemd timer; 6 endpoint probes; $p_{95} < 2$ s).

A dataset-audit run that detects a critical regression *blocks the nightly publish*: the orchestrator (`publish.py`) refuses to advance to the git-push step and instead emits an urgent ntfy alert. Critical conditions include a total-count cliff >1 %, schema break, EGMR re-regression, court not in registry, future date count >50, cross-DB row-count mismatch >0.5 %. Warning conditions (per-court drift >10 % from 7-day median, OCR-quality regression, citation-graph edge drop >5 %) emit alerts but do not block publish.

The framework codifies real incidents. Three case studies illustrate the pattern of *audit detects → fix ships → check codifies*:

Case study 1 — the SG merge anomaly (alphabetical-collision class). On 2026-04-29 a routine total-count check fired: the `sg_publikationen` court had lost 1,450 rows (≈ 90 % loss) on the nightly rebuild without any scraper failure. The diagnostic chain was: (a) the `INSERT OR IGNORE` dedup pattern in `build_fts5.py` processes JSONL shards in alphabetical order; (b) the `entscheidsuche` side-feed (`es_sg_publikationen.jsonl`) processes *before* the direct shard (`sg_publikationen.jsonl`) because 'e' precedes 's'; (c) the two shards happen to construct identical decision-id prefixes (`sg_publikationen_<docket>`); (d) the earlier-processed shard wins all dedups, silently dropping the chamber-correct rows from the direct shard. Reproducer scripts (`scripts/sg_anomaly_*.py`) confirmed the hypothesis on a fresh database. Two fixes shipped on 2026-04-30: (i) the problematic `es_sg_publikationen` feed was retired (commit `f249f1f`); (ii) an architectural fix forced direct shards to process before `es_*` shards (commit `713afe3`, gated by `BUILD_FTS5_DIRECT_FIRST=1`). The pattern check across all multi-chamber cantons confirmed only SG was affected. The fix added two permanent regression checks to the L3 audit: per-court row-count drift >10 % (warning) and per-court row-count drop >50 % (critical, blocks publish).

Case study 2 — the EGMR re-regression class (cross-source double-coverage). The `entscheidsuche CH_BGE` feed includes `cedh` (ECHR) decisions that should land under our `bge_egmr` court rather than `bge`. Without an override, 474 ECHR cases were duplicated with divergent metadata (BGE side carried the full text but no chamber labels, the BGE-EGMR side carried chamber and regeste but truncated text). External audit on 2026-04-29 surfaced this; we backfilled production with a best-of-both merge (preserve full text from BGE side, preserve chamber and regeste from BGE-EGMR side) and deleted the BGE side. The L3 audit now includes a hard-zero check on (`court='bge'` AND `source_url` contains `cedh`): *any* positive count blocks the next publish.

Case study 3 — the König P-class incidents. A second external audit (König, 2026-04-29 to 2026-05-01) uncovered seven distinct issue classes over two days: oversized HUDOC regestes (P1, full judgment text in the regeste field, max 165k chars; truncation pass added in `build_fts5._truncate_oversized_regestes`); short-text fragments that belonged in the regeste field (P7, migration pass added); internal-newline dockets (5,441 rows fixed via extended `_normalize_dockets`); BS/GL relative source URLs (694 rows, host prefix recovery via `_normalize_source_urls`); and ZH targeted gap-fill of 11 missing dockets. Each issue class is now a permanent `check_*` function in `quality/checks/*.py` and a build-time auto-correction pass in `build_fts5.py`.

Detection cadence. During the codification period (2026-04-29 to 2026-05-01), the L3 audit fired at least one critical condition on the majority of nightly runs because each external-audit finding initially manifested as a publish-blocking regression. After each issue class was codified into a build-time auto-correction pass in `build_fts5.py`, the same regression class was prevented at build time on subsequent runs, so the audit no longer fires for it. The drift band (±5×MAD on a 7-day rolling median) flags court-level anomalies typically within one nightly run of the underlying scrape change; a deeper longitudinal study of detection latency is left for future work.

10 Discussion and generalisation

The cross-lingual results have a clear takeaway for legal-RAG practitioners: *retrieval quality is asymmetric across languages and direction.* A naive deployment that only validates same-language MRR will overestimate user-facing quality in mixed-language jurisdictions. Our public benchmark allows quantitative pre-deployment audits of cross-lingual performance.

The verification framework generalises beyond Switzerland. Each rail of the closing audit is corpus-agnostic given (a) an enumerable decision corpus, (b) a statute mirror, and (c) an extraction grammar for the local citation conventions. We sketch an English-law application in Appendix B. The Materialien layer is the most directly Switzerland-specific contribution: jurisdictions with comparable legislative-history doctrine (Germany's *Gesetzgebungsmaterialien*, Canada's *Hansard*, the United States' *legislative history*) could replicate the parliament-chain SPARQL traversal pattern over their own gazette ontologies.

The early-adoption signal — a Swiss-French law firm and a cantonal-government legal department actively querying the deployed MCP within weeks of soft launch — suggests that the resource fills a deployment-blocker gap rather than only a research-evaluation gap. The corollary risk: any infrastructure regression now affects production legal research, which motivates the four-layer QC framework (Section 9) as a non-negotiable accompaniment to release.

11 Limitations and ethics

Benchmark scope. The cross-lingual question set is single-curator and limited to 50 cases (150 trials). It is anchored on the citation graph (target decisions are objectively-ranked leading cases) but the queries themselves are programmatic keyword extractions from the corresponding language's regeste, not free legal-research paraphrases. v2.0 of the benchmark will recruit multi-annotator panels to reach inter-annotator agreement.

Query–target coupling (upper-bound caveat). Because each query is derived from the same case's official multilingual regeste, the query and target share distinctive vocabulary the retrieval system can latch onto. A real legal-research query, written without sight of the regeste, may not contain those discriminators. We therefore read the absolute MRR/Hit@10 numbers as an *upper bound* on realistic retrieval performance, and the cross-cell ranking (which cells are easier or harder, by how much) as the more reliable signal. v2.0 will replace regeste-derived queries with held-out, lawyer-authored research questions.

Italian-target coverage. No top-50 case is originally Italian (top-50 is dominated by social-insurance and procedural BGEs that originate in the German-language stream). This is consistent with the corpus distribution among most-cited decisions but is a coverage gap; v2.0 will explicitly include Italian-original cases sampled at lower citation centrality.

Privacy. Court decisions are public; published BGE/cantonal decisions are typically pseudonymised. We do not republish parties' private identifying information beyond what the courts themselves publish.

Calibration. The grounding-judge rail (R5) is a judge-mediated LLM call; we report a 41.7 % false-positive rate on correct drafts in our v2 paper. Full-binary deployment would over-flag; we recommend the rail be used in advisory mode pending v2.0 calibration work.

12 Open access and artifacts

All artifacts are publicly available:

- Corpus: Hugging Face `https://huggingface.co/datasets/voilaj/swiss-caselaw` (CC-BY-4.0)
- Benchmark: Hugging Face `https://huggingface.co/datasets/voilaj/swiss-legal-rag-bench` (CC0)
- Code: `https://github.com/jonashertner/caselaw-repo-1` (MIT)
- Live MCP: `https://mcp.opencaselaw.ch` (SSE; OpenAPI at `/api/openapi.copilot.json`)
- Word add-in (lawyer-facing client): `https://opencaselaw.ch/word`
- Dashboards: `https://opencaselaw.ch`, `https://opencaselaw.ch/quality.html`

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A Reproducibility manifest

All claims in this paper are backed by code, data, and configuration that anyone can clone or query. This appendix lists every claim's provenance so a reviewer can verify in less than five minutes.

Code.

- **Repository**: `https://github.com/jonashertner/caselaw-repo-1` (MIT licence). The release tag for this paper is `paper-resource-2026-05` (to be assigned at submission; prior to that, all numerical claims trace to the commit pinned in the HuggingFace dataset's `manifest.json`).

- **Five-rail audit pipeline**: `mcp_server.py::_handle_attest_response` (entry at line 9613 of the release tag). The four deterministic rail implementations are inline in the same file: `_audit_statutes` (R2, line 9013), `_audit_quotes` (R3, line 9088), `_audit_dates` (R4, line 9179), `_audit_grounding` (R5, line 9492). R1 (case-citation existence) is inline in `_handle_attest_response`.
- **Quality framework**: `quality/` package (`runner.py`, `cli.py`, `drift.py`, `baseline.py`, `smoke.py`, plus 17 modules in `quality/checks/` containing 61 `check_*` functions).
- **Cross-lingual harness**: `benchmarks/run_cross_lingual.py`.
- **Pipeline guards**: `build_fts5.py` `_normalize_*` / `_dedup_*` / `_truncate_*` passes.

Data artifacts.

- **Corpus**: HuggingFace `voilaj/swiss-caselaw` (CC-BY-4.0). Each release is accompanied by a `manifest.json` listing per-court row counts plus SHA-256.
- **Cross-lingual benchmark v1**: HuggingFace `voilaj/swiss-legal-rag-bench/cross_lingual_v1.js` (CC0). 150 queries, anchored on the citation graph.
- **Audit-bench v2**: same repository, `verification_seed_v2.jsonl`.
- **Citation graph**: derived from the corpus via `search_stack/build_reference_graph.py`; release snapshot available on Zenodo (DOI to be assigned at submission).
- **Statute mirrors**: `statutes.db` (federal, derived from Fedlex SPARQL endpoint), `cantonal_laws.db` (cantonal, derived from LexWork+SIL+ZH portals via `scrapers/cantonal_laws/`).

Live system (runtime verification).

- `https://mcp.opencaselaw.ch/sse` (MCP server endpoint).
- `https://opencaselaw.ch/` (corpus dashboard, multilingual).
- `https://opencaselaw.ch/quality.html` (live dataset health).
- `https://opencaselaw.ch/coverage/` (per-source scrape health).

Reproducing key numerical claims.

```
git clone https://github.com/jonashertner/caselaw-repo-1
cd caselaw-repo-1
git checkout paper-resource-2026-05
pip install -r requirements.txt
huggingface-cli download voilaj/swiss-caselaw --local-dir output/
python -m quality.cli run                  # 4-layer dataset health
python benchmarks/run_search_benchmark.py    # Tab.~2 baseline
python benchmarks/run_cross_lingual.py       # Tab.~3, Fig.~2 (cross-lang)
python -m benchmarks.citation_resolution_analysis    # Sec.~4 unresolved breakdown
```

Telemetry, costs, latency. The MCP server exposes `/health` (corpus row count) and is monitored every 5 minutes by a systemd timer that probes 6 endpoints (search, decision, export docx/pdf/bib/ris) with $p_{95} < 2\,\mathrm{s}$. Every audit run since 2026-05-01 is logged with full per-rail latency and judge cost to `logs/audit_telemetry.jsonl`.

Hardware. Production runs on a Hetzner CCX43 dedicated instance (16 vCPUs, 64 GB RAM, 200 GB attached volume). The full nightly publish takes 3 h 50 min, dominated by the FTS5 rebuild.

Licences.

- Code: MIT.
- Corpus + benchmark: CC-BY-4.0 (corpus) / CC0 (benchmark).
- Commentaries: CC-BY-4.0 (OnlineKommentar.ch) / CC-BY-SA-4.0 (OpenLegalCommentary.ch).
- Federal statutes: derived from Fedlex SPARQL endpoint; Swiss Confederation copyright applies, fair-use distribution permitted.

B English-law generalisation sketch

Goal. Demonstrate that the five-rail closing-audit decomposition is corpus-agnostic, given (a) an enumerable decision corpus, (b) a statute mirror, and (c) a citation extraction grammar for the local conventions. We sketch this for English law (UK + Welsh decisions, BAILII).

Components. **Citation grammar.** English neutral citations follow a fixed schema: `[YEAR] COURT NN`, where `COURT` is one of a small controlled vocabulary (`UKSC`, `EWCA Civ`, `EWCA Crim`, `EWHC`, `UKHL`, `UKPC`, `EWFC`, `EWCOP`, etc.). A regular expression captures these reliably:

```
r"\[(\d{4})\]\s+(UKSC|UKHL|UKPC|EWCA\s+Civ|EWCA\s+Crim
   |EWHC|EWCOP|EWFC|EWCC|UKEAT|UKUT|UKFTT)\s+(\d+)"
```

Statute mirror. `legislation.gov.uk` publishes machine- readable XML for every UK Public General Act since 1267. The R1 statute- existence rail can be implemented identically to our Swiss Fedlex mirror, swapping the `statutes.db` schema in place.

Decision corpus. BAILII publishes UK and Irish decisions free of charge (single-source: subscription model retired, donate-as-you-go). The corpus is approximately 750,000 decisions across UK courts. A single-script crawler is feasible.

What works without modification.

- **R1 case-citation existence**: one regex change (above).
- **R3 verbatim-quote source matching**: identical algorithm, re-uses our 5-gram shingle hashing pipeline.
- **R4 decision-date sanity**: identical (date $\leq$ today and $\geq$ corpus floor).
- **R5 grounding judge**: identical (LLM-mediated, prompt templates have only the corpus name in them).

What requires adaptation.

- **R2 statute resolution**: needs the `legislation.gov.uk` URN scheme rather than Fedlex SR numbers. URN structure is well-documented; estimated 1–2 weeks of integration work.
- **Cross-lingual**: not applicable (English law is monolingual). The cross-lingual benchmark would target a different multilingual jurisdiction (Belgium, Canada-Quebec, South Africa, India).

Status. The architecture above is a sketch only; we have not yet performed an English-law evaluation. Full implementation and evaluation are left for future work. The two implementation tasks estimated above (citation-grammar swap, statute-mirror swap) are the non-trivial portions; the corpus crawl and rail orchestration reuse this paper's harness directly.

C Cross-lingual question set

Schema. Each row is a leading case (anchored on the citation graph) with a parallel query authored in each of DE/FR/IT. The full release at `benchmarks/swiss_legal_rag_bench/cross_lingual_v1.jso` also carries: `q_id`, `target_lang`, `anchor_terms`, `difficulty`.

Table 10: Cross-lingual question set (page 1 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 125 V 351	accident_insurance	DE	Unfallversicherungsgesetz Verbindung VwVG Beweiswürdigung Parteigutachten
BGE 125 V 351	accident_insurance	FR	LAA liaison appréciation preuves expertise
BGE 125 V 351	accident_insurance	IT	LAINF Cost relazione valutazione prove
BGE 134 V 231	accident_insurance	DE	Unfallversicherungsgesetz Beweiswert diagnostischer Methoden funktionelle
BGE 134 V 231	accident_insurance	FR	LAA valeur probante méthodes diagnostiques
BGE 134 V 231	accident_insurance	IT	LAINF valore probatorio metodi diagnostici
BGE 122 V 157	constitution	DE	Bundesverfassung umfassen formellen Anspruch Beizug
BGE 122 V 157	constitution	FR	Cst. permettent déduire droit caractère
BGE 122 V 157	constitution	IT	Cost. Cost essere dedotto diritto
BGE 130 V 343	general_insurance_law	DE	ATSG seit Januar geltenden Fassung
BGE 130 V 343	general_insurance_law	FR	LPGA teneur vigueur janvier relation
BGE 130 V 343	general_insurance_law	IT	LPGA versione vigore gennaio relazione
BGE 138 III 374	civil_procedure	DE	Zivilprozessordnung Beweiserhebung Berufungsverfahren Begründung Berufung
BGE 138 III 374	civil_procedure	FR	CPC administration preuves appel motivation
BGE 138 III 374	civil_procedure	IT	CPC assunzione prove appello motivazione
BGE 133 II 249	federal_court_procedure	DE	Bundesgerichtsgesetz öffentlich-rechtlichen Angelegenheiten Baubewilligung Nachbarbeschwerde
BGE 133 II 249	federal_court_procedure	FR	LTF relation matière droit public
BGE 133 II 249	federal_court_procedure	IT	LTF relazione materia diritto pubblico
BGE 125 V 256	disability_insurance	DE	Invalidenversicherungsgesetz Verbindung Massgebender Hilflosigkeitsgrad Veränderungen
BGE 125 V 256	disability_insurance	FR	LAI relation Degré impotence déterminant
BGE 125 V 256	disability_insurance	IT	LAI relazione Grado grande invalidità
BGE 141 V 281	general_insurance_law	DE	ATSG Verbindung insbesondere psychosomatische Leiden
BGE 141 V 281	general_insurance_law	FR	LPGA corrélation particulier affections psychosomatiques
BGE 141 V 281	general_insurance_law	IT	LPGA relazione particolare affezioni psicosomatiche
BGE 126 V 75	disability_insurance	DE	Invalidenversicherungsgesetz Kürzung Tabellenlöhnen Bestimmung Invalideneinkommens
BGE 126 V 75	disability_insurance	FR	LAI Réduction salaires ressortant statistiques
BGE 126 V 75	disability_insurance	IT	LAI Riduzione salari statistici fini
BGE 133 V 108	disability_insurance	DE	Invalidenversicherungsgesetz Kraft gewesen Dezember Massgebende
BGE 133	disability_insurance	FR	LAI vigueur jusqu décembre Base

Table 11: Cross-lingual question set (page 2 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 126 V 353	accident_insurance	DE	Unfallversicherungsgesetz Arbeitswegunfall Teilzeitbeschäftigten Frage Arbeitspensum
BGE 126 V 353	accident_insurance	FR	LAA Accident trajet victime personne
BGE 126 V 353	accident_insurance	IT	LAINF Infortunio tragitto vittima persona
BGE 115 V 133	accident_insurance	DE	Unfallversicherungsgesetz Präzisierung veröffentlichten adäquaten Kausalzusammenhang
BGE 115 V 133	accident_insurance	FR	LAA Précision apportée publiée relative
BGE 115 V 133	accident_insurance	IT	LAINF Precisazione concernente nesso causalità
BGE 125 V 413	disability_insurance	DE	Invalidenversicherungsgesetz VwVG Verbindung Streitgegenstand Begriff
BGE 125 V 413	disability_insurance	FR	LAI relation objet litige Notion
BGE 125 V 413	disability_insurance	IT	LAI relazione oggetto lite Nozione
BGE 125 V 193	unemployment_insurance	DE	Arbeitslosenversicherungsgesetz Einstellung Anspruchsberechtigung Verlust Anspruchs
BGE 125 V 193	unemployment_insurance	FR	LACI suspension déchéance droit indemnité
BGE 125 V 193	unemployment_insurance	IT	LADI sospensione decadenza diritto indennità
BGE 136 I 229	federal_court_procedure	DE	Bundesgerichtsgesetz Anfechtung Prüfungsergebnisses hier Masterabschluss
BGE 136 I 229	federal_court_procedure	FR	LTF contre résultat examen espèce
BGE 136 I 229	federal_court_procedure	IT	LTF Cost contro esito esame
BGE 130 III 321	civil_code	DE	Zivilgesetzbuch Eintritt Versicherungsfalls Beweis Beweislast
BGE 130 III 321	civil_code	FR	CC survenance sinistre preuve Fardeau
BGE 130 III 321	civil_code	IT	CC sopravvenienza sinistro prova Onere
BGE 132 V 93	general_insurance_law	DE	ATSG VwVG Begutachtung durch Sozialversicherer
BGE 132 V 93	general_insurance_law	FR	LPGA Expertise ordonnée assureur social
BGE 132 V 93	general_insurance_law	IT	LPGA Perizia ordinata dall assicuratore
BGE 134 II 244	federal_court_procedure	DE	Bundesgerichtsgesetz Nichteintreten ungenügend begründete setzt
BGE 134 II 244	federal_court_procedure	FR	LTF Irrecevabilité insuffisamment motivé exige
BGE 134 II 244	federal_court_procedure	IT	LTF Inammissibilità motivato modo sufficiente
BGE 134 I 140	echr	DE	EMRK öffentlich-rechtlichen Angelegenheiten Schutzmassnahmen gegen
BGE 134 I 140	echr	FR	CEDH zurichoise protection contre violence
BGE 134 I 140	echr	IT	CEDU segg Cost legge zurighese
BGE 127 I 38	criminal_procedure	DE	Strafprozessordnung UNO-Pakt Unschuldsvermutung Beschränkung Kognition
BGE 127	criminal_procedure	FR	CPP aCst Pacte présomption innocence

Table 12: Cross-lingual question set (page 3 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 140 III 86	federal_court_procedure	DE	Bundesgerichtsgesetz Pflicht Begründung Rechtsverletzungen Willenserklärung
BGE 140 III 86	federal_court_procedure	FR	LTF obligation motiver violations droit
BGE 140 III 86	federal_court_procedure	IT	LTF obbligo motivare violazioni diritto
BGE 130 V 445	general_insurance_law	DE	ATSG Intertemporalrechtliche Anwendbarkeit materieller Bestimmungen
BGE 130 V 445	general_insurance_law	FR	LPGA Application intertemporelle dispositions matérielles
BGE 130 V 445	general_insurance_law	IT	LPGA Applicabilità intertemporale disposizioni materiali
BGE 137 I 195	constitution	DE	Bundesverfassung Anspruch rechtliches Gehör Replikrecht
BGE 137 I 195	constitution	FR	Cst. droit être entendu réplique
BGE 137 I 195	constitution	IT	Cost. Cost diritto essere sentito
BGE 140 III 115	international_private_law	DE	IPR-Gesetz zweiter Spiegelstrich LugÜ Internationale
BGE 140 III 115	international_private_law	FR	LDIP second tiret Compétence internationale
BGE 140 III 115	international_private_law	IT	LDIP secondo trattino CLug Competenza
BGE 129 I 8	constitution	DE	Bundesverfassung Genfer Reglements April über
BGE 129 I 8	constitution	FR	Cst. Règlement genevois avril fixant
BGE 129 I 8	constitution	IT	Cost. Cost Regolamento ginevrino aprile
BGE 143 IV 241	criminal_procedure	DE	Strafprozessordnung StPO häusliche Gewalt Einstellung
BGE 143 IV 241	criminal_procedure	FR	CPP violence domestique classement pénale
BGE 143 IV 241	criminal_procedure	IT	CPP violenza domestica abbandono procedimento
BGE 141 IV 61	criminal_code	DE	Strafgesetzbuch StGB Mord Konkurrenz mehreren
BGE 141 IV 61	criminal_code	FR	CP assassinat concours entre assassinats
BGE 141 IV 61	criminal_code	IT	CP assassinio concorso reati Riepilogo
BGE 142 II 218	constitution	DE	Bundesverfassung StAhiG VwVG Amtshilfe Steuersachen
BGE 142 II 218	constitution	FR	Cst. assistance administrative matière fiscale
BGE 142 II 218	constitution	IT	Cost. Cost assistenza amministrativa materia
BGE 126 V 319	health_insurance	DE	Krankenversicherungsgesetz Behandlung unfallbedingter Schädigungen Kausystems
BGE 126 V 319	health_insurance	FR	LAMal Traitement lésions système mastication
BGE 126 V 319	health_insurance	IT	LAMal Trattamento lesioni sistema masticatorio
BGE 132 V 393	old_court_law	DE	OG seit Juli geltenden Fassung
BGE 132	old_court_law	FR	OJ teneur vigueur depuis juillet

Table 13: Cross-lingual question set (page 4 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 144 V 210	misc	DE	misc Verordnung Europäischen Parlaments Rates
BGE 144 V 210	misc	FR	misc Règlement Parlement européen Conseil
BGE 144 V 210	misc	IT	misc Regolamento Parlamento europeo Consiglio
BGE 136 IV 55	criminal_code	DE	Strafgesetzbuch StGB Strafzumessung verminderter Schuldfähigkeit
BGE 136 IV 55	criminal_code	FR	CP fixation peine responsabilité restreinte
BGE 136 IV 55	criminal_code	IT	CP commisurazione pena caso scemata
BGE 142 III 413	civil_procedure	DE	Zivilprozessordnung Novenrecht Berufungsinstanz Grundsätze Berufungsverfahrens
BGE 142 III 413	civil_procedure	FR	CPC recevabilité nova appel Principes
BGE 142 III 413	civil_procedure	IT	CPC nova sede appello Principi
BGE 110 V 48	misc	DE	misc Anfechtungsgegenstand Streitgegenstand Verfügung Sachurteilsvoraussetzung
BGE 110 V 48	misc	FR	misc Objet contestation litige existence
BGE 110 V 48	misc	IT	misc Oggetto impugnato litigioso esistenza
BGE 137 III 617	civil_procedure	DE	Zivilprozessordnung Berufungsanträge Berufungseingabe Anträge enthalten
BGE 137 III 617	civil_procedure	FR	CPC conclusions appel mémoire doit
BGE 137 III 617	civil_procedure	IT	CPC conclusioni appello allegato deve
BGE 129 V 1	old_age_insurance	DE	AHVG Ende gültig gewesenen Fassung
BGE 129 V 1	old_age_insurance	FR	LAVS teneur vigueur jusqu corrélation
BGE 129 V 1	old_age_insurance	IT	LAVS tenore vigente fino dicembre
BGE 126 I 97	criminal_code	DE	Strafgesetzbuch BStP Zulässigkeit Geschädigten gegen
BGE 126 I 97	criminal_code	FR	CP Recevabilité droit public interjeté
BGE 126 I 97	criminal_code	IT	CP Ammissibilità diritto pubblico inoltrato
BGE 143 III 65	civil_code	DE	Zivilgesetzbuch Anhörung Pflegeeltern Entlassung Beiständin
BGE 143 III 65	civil_code	FR	CC audition parents nourriciers libération
BGE 143 III 65	civil_code	IT	CC audizione genitori affilianti dimissione
BGE 121 V 362	unemployment_insurance	DE	Arbeitslosenversicherungsgesetz Stellung Personals öffentlicher Dienste
BGE 121 V 362	unemployment_insurance	FR	LACI Statut personnel services publics
BGE 121 V 362	unemployment_insurance	IT	LADI Statuto personale servizi pubblici
BGE 134 IV 1	criminal_code	DE	Strafgesetzbuch Bedingte teilbedingte Strafe gemäss
BGE 134	criminal_code	FR	CP Sursis partiel exécution peine

Table 14: Cross-lingual question set (page 5 of 5). Order: descending in-degree.

Case	Area	Lang	Query
BGE 144 IV 345	criminal_procedure	DE	Strafprozessordnung StPO Unschuldsvermutung Beweiswürdigung Wiederholung
BGE 144 IV 345	criminal_procedure	FR	CPP présomption innocence appréciation preuves
BGE 144 IV 345	criminal_procedure	IT	CPP Cost presunzione innocenza valutazione
BGE 128 III 411	civil_code	DE	Zivilgesetzbuch Kinderunterhaltsbeitrag Untersuchungsmaxime Tragweite Diese
BGE 128 III 411	civil_code	FR	CC contribution entretien faveur enfants
BGE 128 III 411	civil_code	IT	CC contributo mantenimento prole massima
BGE 146 IV 88	criminal_procedure	DE	Strafprozessordnung Verweigerung Massnahmen Feststellung Fahrunfähigkeit
BGE 146 IV 88	criminal_procedure	FR	CPP lien refus mesures constatation
BGE 146 IV 88	criminal_procedure	IT	CPP unitamente LCStr rifiuto sottoporsi
BGE 144 III 349	civil_procedure	DE	Zivilprozessordnung Zulässigkeit Noven Berufungsverfahren uneingeschränkte
BGE 144 III 349	civil_procedure	FR	CPC recevabilité nova appel maxime
BGE 144 III 349	civil_procedure	IT	CPC ammissibilità nova appello principio
BGE 129 I 129	old_court_law	DE	OG Anspruch unentgeltlichen Rechtsbeistand anfechtbarer
BGE 129 I 129	old_court_law	FR	OJ droit être assisté gratuitement
BGE 129 I 129	old_court_law	IT	OG Cost diritto avvocato ufficio
BGE 142 I 135	foreigners_law	DE	AuG Zulässigkeit öffentlich-rechtlichen Angelegenheiten gegen
BGE 142 I 135	foreigners_law	FR	AuG LEtr admissibilité matière droit
BGE 142 I 135	foreigners_law	IT	AuG Cost LStr ammissibilità materia
BGE 138 I 232	contract_tort	DE	Obligationenrecht Personalreglement Öffentlichen Verkehrsbetriebe Genf
BGE 138 I 232	contract_tort	FR	CO Statut personnel Transports publics
BGE 138 I 232	contract_tort	IT	CO Cost segg Statuto personale
BGE 130 II 281	echr	DE	EMRK Familiennachzug gefestigtes Anwesenheitsrecht kombinierten
BGE 130 II 281	echr	FR	CEDH regroupement familial droit présence
BGE 130 II 281	echr	IT	CEDU ricongiungimento familiare diritto residenza
BGE 142 IV 137	misc	DE	misc qualifiziert grobe Verkehrsregelverletzung besonders
BGE 142 IV 137	misc	FR	misc violation grave qualifiée règles
BGE 142 IV 137	misc	IT	misc LCStr grave infrazione qualificata
BGE 123 V 150	unemployment_insurance	DE	Arbeitslosenversicherungsgesetz Verwaltungspraxis wonach Versicherte unwahren
BGE 123	unemployment_insurance	FR	LACI Constitue abus pouvoir appréciation